

Trig Definitions

SYNOPSIS

In the last note, we resolved a vector into a right triangle: the vector itself is the hypotenuse, and its x- and y-components are the two legs. That gave us the picture — $\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y$. What it didn't give us were numbers. SOH–CAH–TOA is what turns that picture into arithmetic: sine, cosine, and tangent are just fixed ratios between a right triangle's *opposite*, *adjacent*, and *hypotenuse* sides. Once we know which side is which, $F_x = F \cos\theta$ and $F_y = F \sin\theta$ fall right out — and running the same idea backward, we can recombine two vectors' components into a single resultant's magnitude and direction. The one catch: the calculator only knows the *angle* you give it, not which way the vector actually points. You have to supply the sign yourself.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 State SOH–CAH–TOA** and identify a right triangle's opposite, adjacent, and hypotenuse sides relative to a given angle θ .
- PE2 Compute a vector's components** using $F_x = F \cos\theta$ (adjacent) and $F_y = F \sin\theta$ (opposite).
- PE3 Assign the correct sign to each component** based on which quadrant the vector actually points into — a reference angle alone doesn't tell you the sign.
- PE4 Recombine components into a resultant** — magnitude $|C| = \sqrt{C_x^2 + C_y^2}$ and direction $\theta = \tan^{-1}(\dots)$ — from two vectors' F_x/F_y values.



You already know this

SOH–CAH–TOA isn't new — you've used right-triangle ratios without the vector dressing:



A leaning ladder



TV screen size



A skate ramp

The ladder is the hypotenuse. How far it reaches up the wall (opposite) versus how far its base sits out (adjacent) depends only on the angle it leans at.

A "55-inch" TV is measured along the diagonal — the hypotenuse. The actual width and height are the adjacent and opposite legs of that same triangle.

Steeper angle, more "opposite" (rise) for the same ramp length (hypotenuse), and less "adjacent" (run). Same triangle, same three ratios.

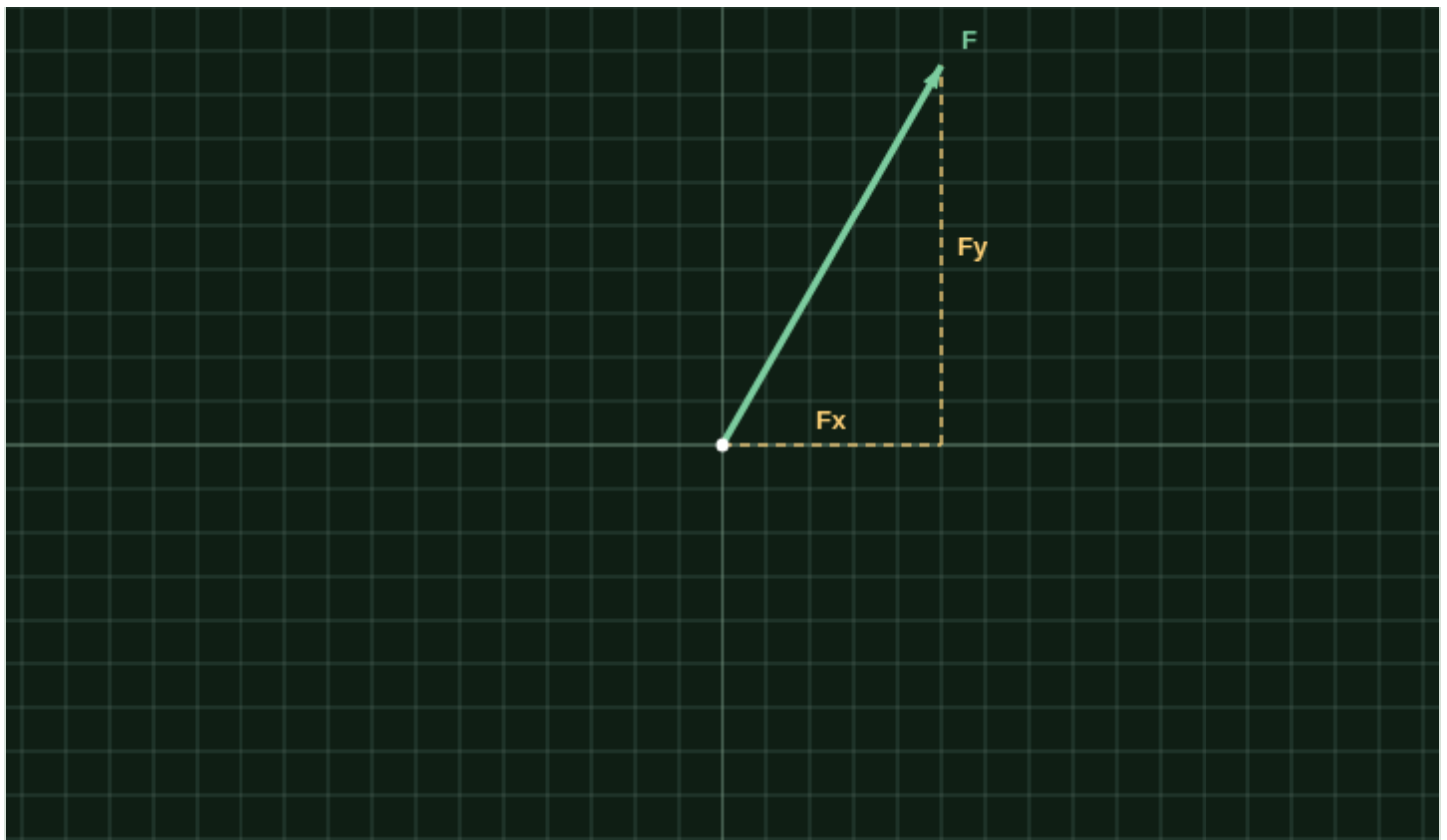
In this note, we work through

- I. What do $\sin\theta$, $\cos\theta$, and $\tan\theta$ mean in terms of a right triangle's sides?
- II. How do we use SOH–CAH–TOA to turn a vector's magnitude and angle into F_x and F_y ?
- III. Why can't we always trust the calculator's sign — how do we decide whether a component is positive or negative?
- IV. Once we have two vectors' components, how do we recombine them into the resultant's magnitude and direction?

Play with it: component calculator

Set a vector's magnitude F and reference angle θ , then pick which quadrant it actually points into. Watch F_x and F_y update live — and notice that the same θ can give a component a plus or a minus sign, depending on the quadrant you chose.

► INTERACTIVE · COMPONENT CALCULATOR



Magnitude F

10.0

Reference angle θ

60°

Quadrant I (+, +)

Quadrant II (−, +)

Quadrant III (−, −)

Quadrant IV (+, −)

$$F_x = F \cos\theta = +5.00$$

$$F_y = F \sin\theta = +8.66$$

The reference angle is always measured from the nearest horizontal axis — the quadrant tells you the sign.

Predict first: a vector's reference angle is 30°, and the diagram shows it pointing down and to the left — into the third quadrant. What are the signs of its x- and y-components?

Both positive

Both negative

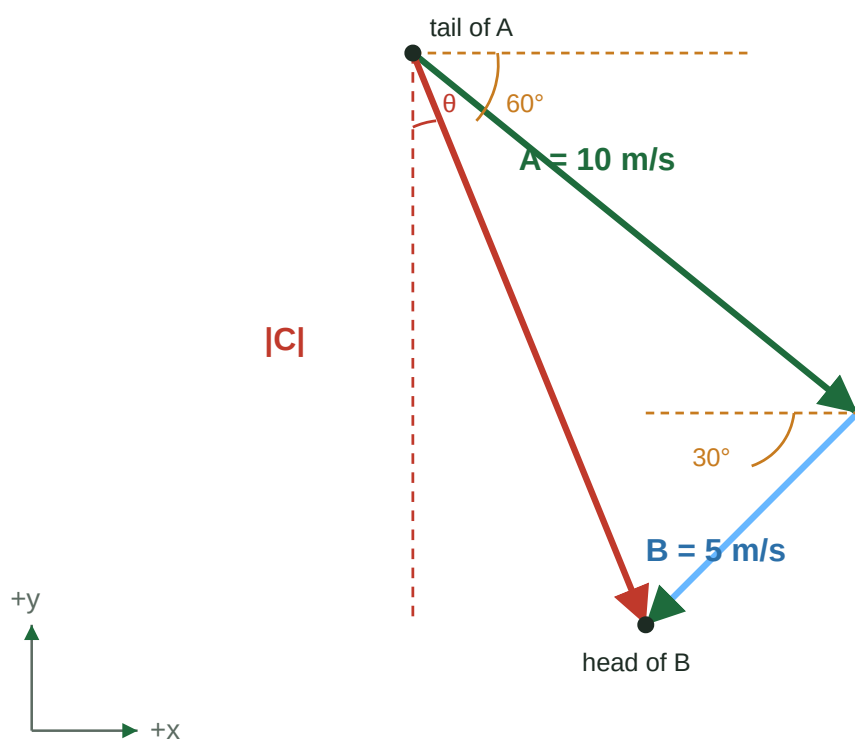
x positive, y negative

x negative, y positive

Worked example: recombining two vectors

Vector **A** (10 m/s) is joined head-to-tail with vector **B** (5 m/s), the same way we joined vectors in Note 1. This time, instead of measuring with a ruler and protractor, we'll find the resultant **C** by the numbers — resolve each vector into components, add the components, then recombine.

Σ WORKED EXAMPLE · **A = 10 M/S ∠60°, B = 5 M/S ∠30°**



STEP 1 · RESOLVE A (QUADRANT IV — RIGHT AND DOWN)

A's reference angle is 60° from the horizontal. Pointing into Quadrant IV, its x-component is positive and its y-component is negative:

$$A_x = +A \cos 60^\circ = +(10)(0.500) = \mathbf{+5.00 \text{ m/s}}$$

$$A_y = -A \sin 60^\circ = -(10)(0.866) = \mathbf{-8.66 \text{ m/s}}$$

STEP 2 · RESOLVE B (QUADRANT III — LEFT AND DOWN)

B's reference angle is 30° from the horizontal — but this time B points into Quadrant III, so *both* components are negative, even though $\cos 30^\circ$ and $\sin 30^\circ$ are themselves positive numbers. The sign comes from the diagram, not the calculator:

$$B_x = -B \cos 30^\circ = -(5)(0.866) = \mathbf{-4.33 \text{ m/s}}$$

$$B_y = -B \sin 30^\circ = -(5)(0.500) = \mathbf{-2.50 \text{ m/s}}$$

STEP 3 • ADD COMPONENTS (THIS IS THE WHOLE REASON TO RESOLVE!)

Components along the same axis add like ordinary numbers — that's the payoff of breaking each vector into x- and y-pieces first:

$$C_x = A_x + B_x = (+5.00) + (-4.33) = \mathbf{+0.67 \text{ m/s}}$$

$$C_y = A_y + B_y = (-8.66) + (-2.50) = \mathbf{-11.16 \text{ m/s}}$$

C_x comes out positive and C_y comes out negative — matching the diagram, where the resultant leans slightly right of straight down.

STEP 4 • RECOMBINE INTO MAGNITUDE AND DIRECTION

Now run SOH–CAH–TOA in reverse. C_x and C_y are the legs of a new right triangle; $|C|$ is its hypotenuse. Because the resultant sits almost straight down, it's most natural here to measure θ from the *vertical* axis instead of the horizontal one — pick whichever axis makes the triangle easiest to see:

$$|C| = \sqrt{(C_x)^2 + (C_y)^2} = \sqrt{(0.67)^2 + 11.16^2} = \mathbf{11.18 \text{ m/s}}$$

$$\theta = \tan^{-1}(|C_x| / |C_y|) = \tan^{-1}(0.67 / 11.16) \approx \mathbf{3.4^\circ}$$

Reading the signs of C_x (+) and C_y (–) tells us the full picture: C points mostly straight down, tilted just 3.4° toward +x.

RESULTANT

C $\approx$ 11.2 m/s, 3.4° from vertical (toward +x)

Watch: working through the note

WALKTHROUGH VIDEO • ~10 MIN

Video embed goes here — print the guided notes below before you press play.

Exit ticket

Four quick questions — check yourself before you close this note.

Q1. In a right triangle with angle θ , $\sin\theta$ equals...

- ☐ adjacent $\div$ hypotenuse
- ☐ opposite $\div$ hypotenuse
- ☐ opposite $\div$ adjacent
- ☐ hypotenuse $\div$ opposite

Q2. When resolving vector F at angle θ , the formula for its component along the horizontal (adjacent) axis is...

- ☐ $F_x = F \sin\theta$

☐ $F_x = F \cos\theta$

☐ $F_y = F \cos\theta$

☐ $F_x = F \tan\theta$

Q3. A vector has magnitude 8.0 m/s at a reference angle of 25° from the horizontal, pointing up and to the right (Quadrant I). What is its y-component, F_y ? (nearest 0.1)

m/s

Q4. A vector's reference angle is 40° , but the diagram shows it pointing down and to the left — into the third quadrant. Its x- and y-components are...

☐ Both positive.

☐ Both negative.

☐ x positive, y negative.

☐ x negative, y positive.

Check my answers

Clear

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

Print or open on a tablet and work the SOH–CAH–TOA / component problems as you follow along.

AFTER THE MODULE

Completed notes (PDF)

Released when Module 2 closes — check your work against it.

